🚀 EduTrack — Deployment Guide

Academic Management Platform

VTU Internship 2026 — Team 15

Overview

EduTrack is deployed across three cloud platforms:

COMPONENT	PLATFORM	URL
Frontend (React)	Vercel	https://vtu-intern-2026-team15-java-fs.vercel.app
Backend (Spring Boot)	Render	https://edutrack-backend-o1ym.onrender.com
Database (MySQL)	Railway	hopper.proxy.rlwy.net:12491

Prerequisites

- GitHub account with the repository pushed
- Railway account (https://railway.app) — sign up with GitHub
- Render account (https://render.com) — sign up with GitHub
- Vercel account (https://vercel.com) — sign up with GitHub
- Gmail account with App Password configured

Part 1 — Database Deployment (Railway)

Step 1 — Create Railway Project

1. Go to https://railway.app
2. Click **New Project**
3. Select **Database** → **MySQL**
4. Railway creates a MySQL 9.4 instance automatically
5. Wait for status to show **Online**

Step 2 — Get Connection Details

1. Click the MySQL service → **Variables** tab
2. Note down:
   - `MYSQL_PUBLIC_URL` — full connection string
   - `MYSQLHOST` — internal hostname

- `MYSQLPASSWORD` — root password
- `MYSQLPORT` — public port

Step 3 — Import Schema

1. Open **MySQL Workbench**
2. Create new connection:
   - Hostname: value from `MYSQL_PUBLIC_URL` (after `@`, before `:`)
   - Port: port number from `MYSQL_PUBLIC_URL`
   - Username: `root`
   - Password: `MYSQLPASSWORD` value
3. Click **Continue Anyway** if version warning appears
4. File → **Open SQL Script** → select `EduTrack-Database/Schema.sql`
5. Click the **lightning bolt** ⚡ to execute
6. All 11 tables are created

Part 2 — Backend Deployment (Render)

Step 1 — Create Dockerfile

The `Dockerfile` is already in `EduTrack-Backend/`:

```
FROM maven:3.9.6-eclipse-temurin-21 AS build
WORKDIR /app
COPY pom.xml .
COPY src ./src
RUN mvn clean package -DskipTests

FROM eclipse-temurin:21-jre
WORKDIR /app
COPY --from=build /app/target/edutrack-backend-1.0.0.jar app.jar
EXPOSE 8080
ENTRYPOINT ["java", "-jar", "app.jar"]
```

Step 2 — Create Web Service on Render

1. Go to https://render.com → **New +** → **Web Service**
2. Connect GitHub → select `VTU_INTERN_2026_Team15_JavaFS`
3. Configure:
   - **Name**: `edutrack-backend`
   - **Language**: `Docker`
   - **Root Directory**: `EduTrack-Backend`
   - **Dockerfile Path**: `./Dockerfile`
   - **Instance Type**: `Free`

Step 3 — Add Environment Variables

In Render → **Environment** tab, add:

KEY	VALUE
SPRING_DATASOURCE_URL	jdbc:mysql://hopper.p useSSL=false&serverTi
SPRING_DATASOURCE_USERNAME	root
SPRING_DATASOURCE_PASSWORD	Railway MySQL password
SPRING_JPA_HIBERNATE_DDL_AUTO	update
SPRING_MAIL_HOST	smtp.gmail.com
SPRING_MAIL_PORT	587
SPRING_MAIL_USERNAME	your Gmail address
SPRING_MAIL_PASSWORD	Gmail App Password (16 cha
SPRING_MAIL_PROPERTIES_MAIL_SMTP_AUTH	true
SPRING_MAIL_PROPERTIES_MAIL_SMTP_STARTTLS_ENABLE	true

Step 4 — Deploy

1. Click **Create Web Service**
2. Render builds the Docker image (takes 5-10 minutes)
3. Watch logs for: `Started EduTrackApplication`
4. Backend URL: `https://edutrack-backend-o1ym.onrender.com`

Step 5 — Verify

Open in browser:

```
https://edutrack-backend-o1ym.onrender.com/api/system/settings
```

Should return:

```
{"id":1,"activeYear":"2025-26","activeSemester":"Even","maintenanceMode":false,"registrationOpen":true}
```

Note: Render free tier sleeps after 15 minutes of inactivity. First request after sleep takes ~50 seconds. Always wake up the backend before a demo by opening the above URL.

Part 3 — Frontend Deployment (Vercel)

Step 1 — Create Project on Vercel

1. Go to https://vercel.com → **Add New** → **Project**
2. Import `VTU_INTERN_2026_Team15_JavaFS` from GitHub
3. Configure:
   - **Root Directory**: `EduTrack-Frontend`
   - **Framework Preset**: `Create React App` (auto-detected)

Step 2 — Add Environment Variables

Before clicking Deploy, add:

KEY	VALUE
`REACT_APP_API_URL`	`https://edutrack-backend-o1ym.onrender.com`
`DISABLE_ESLINT_PLUGIN`	`true`

Step 3 — Deploy

1. Click **Deploy**
2. Vercel builds the React app (takes ~2 minutes)
3. Frontend URL: `https://vtu-intern-2026-team15-java-fs.vercel.app`

Step 4 — Verify

Open the frontend URL in browser and try logging in:

- Email: `admin@edutrack.com`
- Password: `Admin@123`

Part 4 — Gmail App Password Setup

Required for OTP email delivery:

1. Go to https://myaccount.google.com
2. Click **Security**
3. Enable **2-Step Verification** (must be ON)
4. Search **App Passwords**
5. Select app: **Mail**, device: **Other** → type "EduTrack"
6. Click **Generate**
7. Copy the **16-character password**
8. Use this as `SPRING_MAIL_PASSWORD` in Render environment variables

Part 5 — Auto-Deploy on Code Push

Both Render and Vercel are connected to GitHub and auto-deploy on every push to `main`:

```
git add .
git commit -m "your changes"
git push
```

- Vercel rebuilds the frontend automatically (~2 min)
- Render rebuilds the Docker image automatically (~5-10 min)

Part 6 — CORS Configuration

The backend allows requests from any origin for production:

```
// SecurityConfig.java
config.setAllowedOriginPatterns(List.of("*"));
```

All controllers use:

```
@CrossOrigin(origins = "*")
```

This ensures the Vercel frontend can communicate with the Render backend.

Troubleshooting

ISSUE	CAUSE	FIX
"Cannot connect to server"	Backend is sleeping	Open backend URL and wait 50 seconds
"Access denied for user root"	Wrong DB password	Check `SPRING_DATASOURCE_PASSWORD` in Render
Build failed on Render	Java version mismatch	Ensure Dockerfile uses `eclipse-temurin:21`
ESLint errors on Vercel	CI treats warnings as errors	Add `DISABLE_ESLINT_PLUGIN=true` in Vercel env vars
CORS error in browser	Origin not allowed	Check `@CrossOrigin(origins = "*")` on all controllers
OTP email not received	Wrong Gmail App Password	Regenerate App Password and update Render env var

Live URLs

SERVICE	URL
Frontend	https://vtu-intern-2026-team15-java-fs.vercel.app
Backend	https://edutrack-backend-o1ym.onrender.com
Backend Health	https://edutrack-backend-o1ym.onrender.com/api/health/db
GitHub	https://github.com/NaveenPR123/VTU_INTERN_2026_Team15_JavaFS